



Review

Large language model-based task planning for service robots: A review

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ABSTRACT

With the rapid advancement of large language models (LLMs) and robotics, service robots are increasingly becoming an integral part of daily life, offering a wide range of services in complex environments. To deliver these services intelligently and efficiently, robust and accurate task planning capabilities are essential. This paper presents a comprehensive overview of the integration of LLMs into service robotics, with a particular focus on their role in enhancing robotic task planning. First, the development and foundational techniques of LLMs, including pre-training, fine-tuning, retrieval-augmented generation (RAG), and prompt engineering, are reviewed. We then explore the application of LLMs as the cognitive core—"brain"—of service robots, discussing how LLMs contribute to improved autonomy and decision-making. Furthermore, recent advancements in LLM-driven task planning across various input modalities are analyzed, including text, visual, audio, and multimodal inputs. Finally, we summarize key challenges and limitations in current research and propose future directions to advance the task planning capabilities of service robots in complex, unstructured domestic environments. This review aims to serve as a valuable reference for researchers and practitioners in the fields of artificial intelligence and robotics.

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1. Introduction

With the continuous progress of robotics [1,2] and artificial intelligence (AI) [3,4], the perception, reasoning, and action capabilities of service robots have been significantly enhanced. As a result, service robots are evolving into capable assistants within domestic environments [5–7]. By autonomously performing various household tasks such as desktop organizing, floor cleaning and even caring for the elderly, these robots contribute to improving the quality of life [8,9].

To effectively accomplish daily tasks, service robots must engage in detailed task planning, which involves understanding user requirements, decomposing high-level goals into subtasks, and

generating executable action sequences [10]. Task planning is the core technology to realize autonomous decision-making of service robots, which enables the robots to analyze environmental information, evaluate task requirements, and develop optimal action plans. Such autonomous decision-making capability is the key to robot intelligence. Traditional robotic task planning methods mainly rely on detailed task models and environment information [11–13]. For example, the task planning problem is typically divided into the domain description and problem description [14], where the domain describes a set of actions defined by their preconditions and subsequent effects, while the problem description specifies the initial state and the desired goal conditions. However, such approaches generally presuppose finiteness, determinism, and invariance of planning goals. These assumptions limit their applicability, especially in real-world scenarios where environments are dynamic and unpredictable. Moreover, traditional methods often suffer from low fault tolerance, limited scalability, and weak adaptability to real-world interactions. In real-world domestic environments, the positional relationship between objects and scenes is usually dynamically changing [15]. Challenges such as clutter, occlusion between individual objects

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further lead to an increase in the perception difficulty of service robots [16]. In addition, due to the lack of a priori knowledge of the environment and the target object, the efficiency of target search in large-scale scenarios tends to be low, and blind search of the entire home environment is time-consuming, labor-intensive and difficult to apply to real life [17,18]. Therefore, how to generate executable task planning that meets user needs in complex and dynamically changing environments is one of the critical challenges in deploying service robots in unstructured domestic environments.

In recent years, with the continuous development of large language models (LLMs), they have shown great superiority and adaptability in various fields [19–21]. These giant neural networks with tens of billions of parameters have shown unprecedented comprehension and generation capabilities through training on massive amounts of data. From the stunning performance of ChatGPT to the multimodal breakthrough of GPT-4, LLMs are redefining the boundaries of machine intelligence. The deep fusion of LLM and robotics [22] is driving the evolution of robots from single-task performers to general-purpose intelligences. With their rich common-sense knowledge and reasoning ability as the cognitive core of the intelligences, these models have dramatically improved the service robot's ability to understand and adapt to the environment. And through the fusion processing of multimodal data such as vision and speech, the robot is able to more accurately perceive and understand the surrounding environment, providing a reliable basis for autonomous decision-making. LLMs show even more amazing potential in task planning for service robots. They are able to break down abstract instructions into concrete steps, consider environmental constraints, and formulate reasonable action plans. This ability allows the robots to be no longer limited to preset programs, but to respond flexibly to a variety of complex tasks, understand user needs in depth, and even provide personalized services.

Recent literature has seen a surge in reviews exploring the integration of LLMs with robotics. Kim et al. [23] offered a foundational analysis of LLM-enhanced robotic systems, covering communication, perception, and control. Zeng et al. [24] and Li et al. [25] delved into specific applications, respectively examining LLMs' impact on decision-making and path planning, and their versatility in multi-robot coordination. Cui et al. [26] took a unique task-centric approach, investigating autonomous task discovery through a fusion of visual semantics and LLMs. Furthermore, several reviews have focused on narrower, yet critical, subfields. These include surveys on traditional robotics topics like navigation and perception [27–29], as well as those centered on advanced LLM-driven functionalities such as reasoning and tool use [30,31].

While existing surveys have comprehensively covered various aspects of robotics and LLM integration, our review identifies a critical gap in systematic analysis of task planning specifically for domestic service robots. This domain presents unique challenges due to the unstructured nature of home environments and diverse user interaction patterns, which demand specialized approaches not fully addressed in prior works. Unlike previous reviews that broadly survey LLM applications in robotics, our work not only provides a concise and systematic overview of the fundamentals and core techniques of LLMs, emphasizing their application to task planning for service robots operating in home environments, but also pioneers a modality-centric taxonomy (text, vision, audio, multimodal) for analyzing task planning challenges, enabling more targeted technological development. This novel framework reveals previously overlooked intermodal synergies and implementation barriers specific to home service scenarios.

The key innovations and contributions of this paper are as follows:

- We present a systematic review of the development of LLMs and provide a concise overview of their core techniques, which include pre-training, fine-tuning, retrieval-augmented generation, and prompt engineering.
- We establish the first modality-centric taxonomy for analyzing LLM-based task planning in domestic service robotics, thereby revealing unique cross-modal challenges.
- We critically examine the key obstacles hindering real-world deployment of LLM-based service robots and propose actionable research pathways to advance the field.

The rest of the paper is organized as follows. In [Section 2](#), we first introduce the important theoretical foundations of the LLM, and in [Section 3](#) we provide a systematic categorization of LLM-based task planning for home service robots based on the differences in the input modalities, and provide a detailed overview of each category in [Sections 4 to 7](#). In [Section 8](#), we present some challenges to the current LLM-based robotic task planning and an outlook to the future. Finally, we conclude the paper in [Section 9](#).

2. Theory foundation of LLM

In this section, we provide an overview of the theoretical foundations of LLM, covering the developmental background of LLM, the key techniques — Pre-training, Fine-tuning, RAG, and Prompt Engineering. In addition, we provide an overview of the application of LLM to robotic task planning, showing how it empowers robots to achieve more efficient and intelligent task execution.

2.1. Background

Since the proposal of the Transform architecture, the development of LLM has gone through two core breakthrough phases, which have driven the vigorous development of the natural language processing (NLP) field. The first phase is marked by the proposal of Transformer [32], whose self-attention mechanism enables parallelized long sequence modeling. GPT-1 [33,34] adopt unidirectional and bidirectional pre-training, respectively, to lay down the two paradigms of generation and comprehension, and the pre-training-fine-tuning framework achieves significant improvement on several NLP tasks. The second phase focuses on model scale-up and interaction capabilities, with GPT-3 [35] and ChatGPT [36] demonstrating the potential of generative AI and optimized dialog interactions, respectively. Current research trends are evolving towards efficient, autonomous intelligences and ethical alignment, gradually approaching the boundary of general-purpose artificial intelligence (AGI). [Fig. 1](#) illustrates the evolution of LLMs since 2019, and we describe some of these typical LLMs [Table 1](#) below.

T5 [37] improves on the traditional Transformer model by moving layer normalization outside of residual connections. It employs masked language modeling as a pre-training target, using spanwise masking of consecutive tokens instead of independent masking of individual tokens, which shortens sequence length and accelerates training. Once pre-training is complete, an adapter layer [38] is used to fine-tune for downstream tasks.

GPT-3 [35] continues the architecture of GPT-2 [33] and draws on the design of Sparse Transformers [39], combining dense and sparse attention mechanisms in the transformer layer and employing a gradient noise scale during training. GPT-3 extends the model parameters to 175 billion, further validating the positive correlation between model size and performance improvement.

GLaM [40] represents a family of language models employing a sparsely activated Mixed (MoE) structure of decoder experts [41].

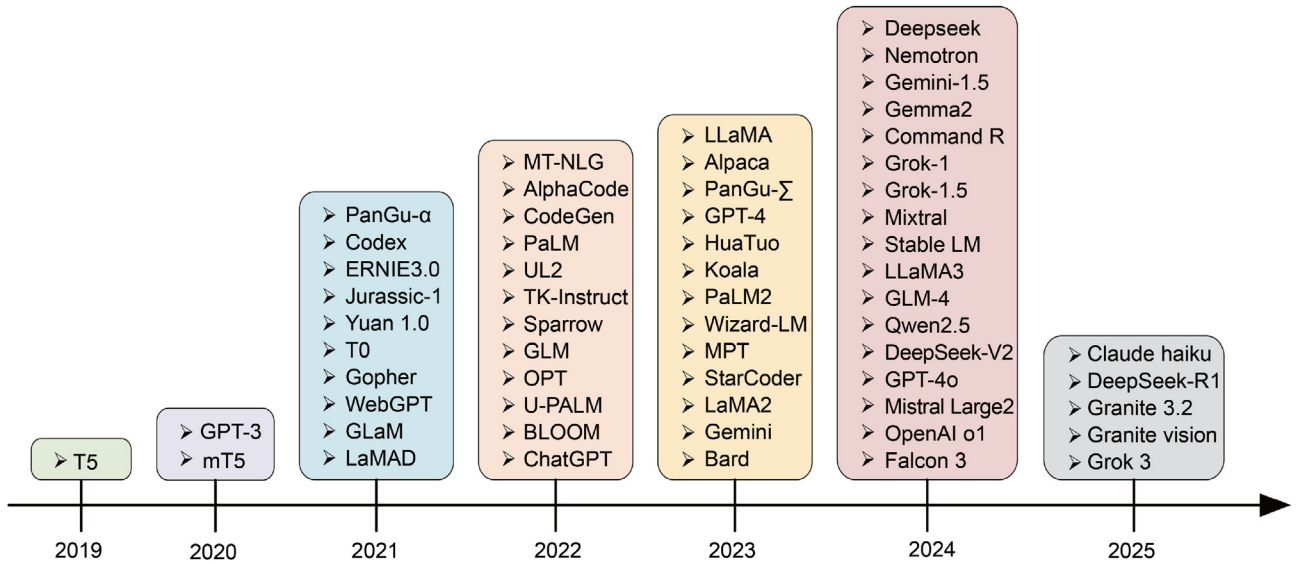


Fig. 1. The evolution of LLM since 2019.

Table 1
Comparison of typical LLMs.

Model	Parameters	Advantages	Application areas
T5 [37]	11B	Moves layer normalization outside of residual connections, uses spanwise masking, and employs an adapter layer for fine-tuning.	Text classification, question answering, summarization.
GPT-3 [35]	175B	Combines dense and sparse attention mechanisms, employs a gradient noise scale, and scales up to 175 billion parameters.	Text generation, language understanding, code generation.
GLaM [40]	1200B	Employs a sparsely activated MoE structure, processes each input token by only the optimal two experts, and reduces training energy consumption.	Text generation, language understanding.
GLM-130B [42]	130B	Uses autoregressive masks for pre-training targets and applies gradient contraction of the embedding layer for training stability.	Bilingual text generation (English–Chinese), language understanding.
DeepSeek-v2 [43]	236B	Introduces Multihead Latent Attention (MLA) to compress the key–value (KV) cache into potential vectors, significantly reducing inference cost.	Efficient text generation, language understanding.

GLaM sparsely activates the experts, with each input token being processed by only the optimal two experts. The model keeps the training energy consumption at one-third of GPT-3 while maintaining high performance.

GLM-130B [42] is a bilingual (English–Chinese) model, which is distinguished from the unidirectional GPT-3 [35] by the use of autoregressive masks to populate the pre-training targets for training and the application of gradient contraction of the embedding layer to ensure training stability.

DeepSeek-v2 [43] is a MoE model that introduces Multihead Latent Attention (MLA) to reduce the inference cost by compressing the key–value (KV) cache into potential vectors. Due to MLA, the inference throughput of DeepSeek-v2 is 5.76 times faster than DeepSeek (see Table 1).

2.2. Pre-training

The pre-training of LLM is the most critical basic stage in the whole model construction process, and this process is essentially a self-supervised learning approach that allows the model to automatically extract linguistic laws, construct a knowledge system, and form a deep understanding of natural language from massive unlabeled text data. Modern LLM pre-training is

mainly built on the Transformer architecture [44], and according to different application scenarios and functional requirements, researchers have developed three main architectural variants: causal decoder architecture, encoder–decoder architecture, and prefix decoder architecture. Although these architectures differ in their specific implementations, they all rely on Transformer's core component, the self-attention mechanism, to achieve effective modeling of long-distance dependencies through techniques such as multi-attention, residual connectivity, and layer normalization.

Pre-training task design is the core innovation of LLM pre-training. Autoregressive language modeling (e.g., GPT) requires the model to predict the next token based on the above, which perfectly fits the text generation and is the classic pre-training paradigm. In addition, improvement schemes such as continuous fragment masking used by SpanBERT [45] and unified language modeling used by UniLM [46] enhance the pre-training effect to varying degrees. In addition the model training phase faces the challenge of optimizing the parameters and data on an ultra-large scale. Researchers have developed optimizers such as AdamW [47] and AdaFactor [48] for large-scale training. The combined use of data parallelism, model parallelism, and pipeline

parallelism substantially improves the training efficiency. Techniques such as mixed-precision training and gradient checkpointing effectively reduce the video memory occupation, making the training process more efficient.

2.3. Fine-tuning

The advantages of fine-tuning are its efficiency and performance improvement. Compared to training from scratch, fine-tuning greatly saves computational resources and time. Pre-trained LLMs are good at predicting textual tokens, but may be limited in generating structured output or processing domain-specific information. To overcome these limitations, the output layer of the LLM can be tuned by Fine-Tuning to fit specific task requirements. To address the lack of knowledge in LLM, fine-tuning training using domain-specific data can enhance LLM's understanding and processing in the domain and improve its accuracy and performance on tasks in the domain.

From the perspective of technical implementation, fine-tuning is mainly divided into two major directions, Full Fine-tuning (FFT) [49] and Parameter-Efficient Fine-tuning (PEFT) [50]. FFT updates all the parameters of the model, which is effective but requires a large amount of computational resources and training data, in contrast, PEFT [38,51] drastically reduces the resource requirements while guaranteeing the performance by adjusting only some of the parameters or by adding small adaptation modules. In addition, Instruction Fine-tuning [52] is an important fine-tuning paradigm that has emerged in recent years, which shapes the behavioral patterns of a model by providing it with explicit examples of instructions. Reinforcement Learning with Human Feedback (RLHF) [53] further enhances the effectiveness of Instruction Fine-tuning by introducing an artificial scoring mechanism to make the model outputs more consistent with human values and preferences.

2.4. Prompt engineering

Prompt Engineering is a key technique in LLM applications that does not require updating the model's parameters and aims to guide the model to generate more accurate and reliable outputs by carefully designing input prompts. A well designed prompt can significantly improve model performance, reduce illusions, and adapt to different task requirements. In the following, we will discuss commonly used prompt methods.

Instruction Prompting [54] aims to provide the LLM with examples of instruction prompts so that it can eliminate training or testing discrepancies and simulate real-world usage scenarios for chatbots. Zero-shot Prompting [55] involves feeding tasks into the model without any examples indicating the desired output, requiring the LLM to answer the user's questions without showing any examples in the prompts. Few-shot Prompting [35] works by providing the model with a small number of high-quality examples that include the inputs and desired outputs of the target task. By observing these good examples, the model can better understand human intent and the criteria for generating accurate output. Chain-of-Thought Prompting [56] generates a series of short sentences, known as chains of reasoning. These sentences describe step-by-step reasoning logic that leads to a final answer, which can be more beneficial for tasks with complex reasoning and larger models. Recursive Prompting [57] is a problem solving methodology that involves breaking down complex problems into smaller, more manageable subproblems and then solving these subproblems recursively through a series of prompts.

2.5. Retrieval-augmented generation

RAG [58] is an important technological breakthrough in the field of natural language processing in recent years, which effectively solves the inherent limitations of traditional LLM in terms of knowledge timeliness, factual accuracy, and interpretability by organically integrating the information retrieval system with the generative capability of LLM. The core innovation of the technology lies in the establishment of a dynamic knowledge injection mechanism that enables the generative model to access and utilize external knowledge bases in real time, thus significantly enhancing the reliability and expertise of the generated content.

RAG usually consists of two main components, Retrieval and Generation. First, the model retrieves a set of relevant documents from a pre-constructed document collection based on an input query. The retrieved documents are then fed into the LLM along with the query, and the model generates the final answer based on these inputs. Some typical RAG approaches include Multi-Head RAG (MRAG) [59], which utilizes multi-head attention to enhance multifaceted problem processing. Adaptive-RAG [60], which dynamically selects strategies to cope with queries of varying complexity. Blended RAG [61], which fuses semantic search and hybrid query strategies to improve accuracy. Self-RAG [62], enhancing model quality and factuality through search and self-reflection. As well as IM-RAG [63], learning introspective monolog integrates LLM and IR systems to support multi-round RAG. These approaches have their own characteristics, and together they promote the development of RAG technology.

2.6. Application of LLMs in robotic task planning

In recent years, LLMs have made significant progress in the field of natural language processing. With the continuous evolution of technology, the application of LLMs has broken through the traditional scope of text generation and analysis, and begun to penetrate into emerging fields such as robot task planning. As shown in Fig. 2, the combination of LLMs and home service robots is driving the development of more intelligent autonomous agents, with LLMs acting as the "brain" of the service robots, which significantly improves their ability in task planning.

When the robot receives ambiguous commands from the user, LLM is able to parse and understand them to accurately capture the user's intent [58,64,65]. For example, in a home service robot scenario, the user may say "put that thing on the high shelf", and LLM understands that "that thing" refers to the water cup on the table, and recognizes that "high" refers to the top shelf of the bookshelf. and recognizes that "high up" refers to the top shelf of the bookcase. Based on this, it combines logical reasoning to determine the optimal execution plan, such as picking up the water cup first, then planning a path to reach the bookshelf avoiding obstacles, and finally placing the water cup at the specified location.

For better task planning, the robot may also rely on VLMs to enhance environment perception [66]. In the above home service scenario, the VLM can recognize the exact location of the water cup, the shape and height of the bookshelf, as well as the obstacles (e.g., chairs or carpets) on the path, providing precise physical information for task planning. Faced with a complex task, the robot is able to break it down into a series of executable sub-tasks, with detailed, robot-executable task planning for each sub-task. For example, a robot can break down "get a glass of water" into steps such as "move to the table" and "grab the glass". The steps of "navigating to the bookshelf", "adjusting the posture", and "placing the cup" are broken down for "placing the cup". In this process, LLMs assisted in detailed planning to ensure that each subtask was properly executed, thus improving the efficiency and accuracy of the overall task planning.

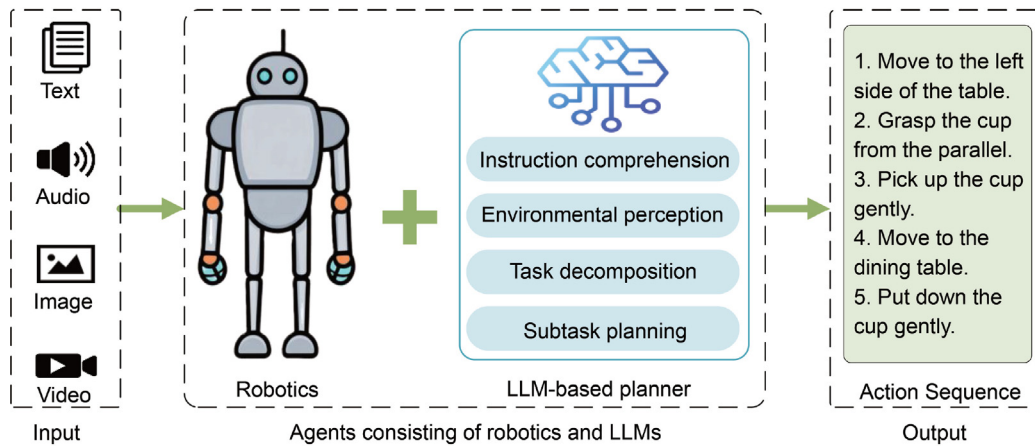


Fig. 2. LLM-based task planning for service robots. The robot acquires information in the form of text, video, audio, etc., and passes the processed input signals to the LLM, which carries out the planning of detailed action sequences through natural language processing, task decomposition, and so on.

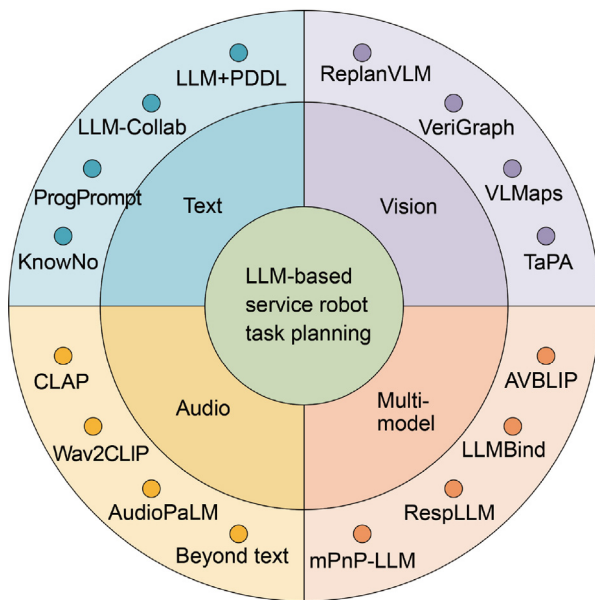


Fig. 3. LLM-based task planning taxonomy for service robots. From the point of view of different input modalities, it is categorized into four categories such as text input, visual input, audio input, and multimodal input.

3. Taxonomy

Against the backdrop of the intelligent development of domestic service robots, task planning systems are evolving from unimodal to multimodal fusion. Based on the differences in input modalities, this paper constructs a systematic classification framework, dividing LLM-driven task planning methods for domestic service robots into four major categories: Text-based LLM Planning, Vision-Language Model Planning, Audio-based Planning, and Multimodal Large Language Models Planning, as illustrated in Fig. 3. This framework elucidates the technical characteristics and performance boundaries under different modality combinations. Here, we briefly summarize these four categories as follows.

Text-based LLM Task Planning. As the most fundamental implementation paradigm, this planning method primarily relies on text-only input for task planning. Such systems typically employ Transformer-based language models as core processors, utilizing natural language understanding (NLU) modules to parse

user instructions and generate corresponding action sequences. However, due to the lack of direct environmental perception capabilities, these systems require pre-constructed semantic maps or knowledge graphs to supplement environmental information.

Vision-based LLM Task Planning. This method significantly enhances robots' understanding of physical environments by integrating visual perception capabilities. VLMs focus primarily on establishing semantic alignment between visual and linguistic modalities. VLMs usually adopt dual-encoder architectures or cross-modal Transformers to realize efficient visual-linguistic interactions, and their typical applications include: VQA, image description generation, visual localization. Through cross-modal attention mechanisms, the system establishes a joint "vision-language" representation space, enabling complex instructions requiring spatial reasoning, such as "place the mug on the dining table into the dishwasher".

Audio-based LLM Task Planning. The system enables the robot to understand and respond to voice commands by introducing an audio processing module. Such systems usually include speech recognition module and speech synthesis module, which can convert user's voice commands into text and generate corresponding action sequences. Through the analysis of audio signals, robots can recognize and respond to specific voice commands in noisy environments, thereby improving their flexibility and adaptability in practical applications.

Multimodal LLM Task Planning. The Multimodal LLM (MLLM) cover not only vision and speech, but also integrate audio, sensor data, robot state, and other modalities to support more complex task planning. MLLMs have three main features: multimodal inputs, dynamic modal fusion, and cross-modal reasoning. Such systems usually adopt a hierarchical fusion strategy: modal alignment is achieved through cross-attention at the feature level, and the contribution weight of each mode is dynamically adjusted by the gating mechanism at the decision level. In this way, the MLLM Planning system can integrate multiple information sources in a complex and changing home environment to generate more accurate and flexible task planning.

4. Text-based LLM task planning

As the basic paradigm of LLM-driven task planning, text-based systems establish a key baseline for human-computer interaction through natural language processing. These systems convert unstructured natural language instructions into executable action sequences by using the semantic understanding ability of the converter architecture, representing the most computationally efficient method in the modal spectrum.

Table 2
Typical literature on text-based LLM task planning.

Paper	Year	Core innovation	Limitations
Ding et al. [67]	2023	Achieves cross-environment object rearrangement by combining semantic knowledge with geometric planning	Requires precise object labeling and limited to tabletop-scale tasks
Singh et al. [65]	2023	Enables generalized planning through programmatic prompts supporting multi-modal specifications	Demands complex prompt engineering with high computational overhead
Silver et al. [58]	2024	Ensures plan validity through direct PDDL integration with formal correctness guarantees	Depends on pre-defined PDDL domains and scales poorly to novel objects
Ren et al. [68]	2023	Provides statistical uncertainty quantification while minimizing human intervention	Tends to produce conservative plans with complex calibration requirements
Wei et al. [56]	2022	Simplifies complex reasoning via chain-of-thought prompting without model modification	Lacks formal verification and sensitive to prompt phrasing variations
Liu et al. [69]	2024	Enables long-horizon planning through autoregressive goal decomposition	Suffers from error accumulation in sub-tasks requiring fine-tuning
Kannan et al. [70]	2024	Facilitates multi-robot collaboration by handling capability heterogeneity	Requires centralized coordination introducing communication overhead

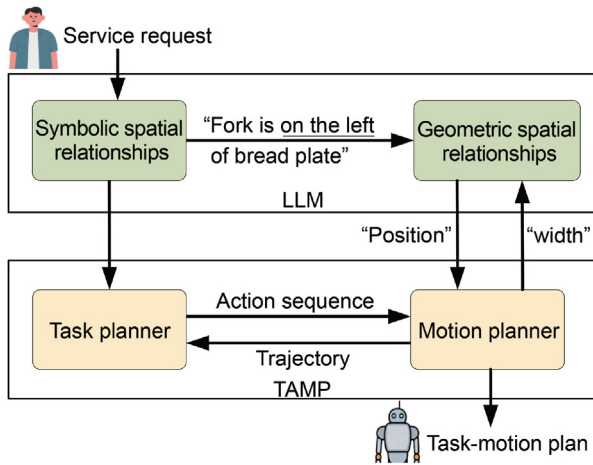


Fig. 4. Schematic diagram of the architecture of LLM-grop. Source: Adapted from [67].

The task planning system of LLM service robot based on text faces two core challenges: one is how to effectively deal with instructions of different complexity (from simple short-term tasks to complex long-term planning), and the other is how to make up for the lack of environmental perception caused by pure text input. To this end, this section will systematically elaborate the existing research results from two dimensions: the planning method based on instruction complexity and the compensation technology of environment-aware defects.

4.1. Planning method based on instruction complexity

Due to the excellent performance of LLM in generalization and common sense reasoning, it can skillfully understand natural language instructions and perform logical reasoning, translation and zero sample planning. This section will introduce Text-based LLM Planning in terms of the complexity of instructions, and some of the representative work is shown in Table 2. In the face of simple short-term tasks, LLM shows excellent planning ability. The LLM-grop proposed by Ding et al. [67] extracts common sense knowledge about semantically valid object configuration from LLM through the prompt function, and instantiates them with tasks and motion planners in order to promote them to different scene geometries, as shown in Fig. 4. LLM-grop can achieve object rearrangement from natural language commands to human alignment in various environments. ProgPrompt, proposed by Singh et al. [65], is a programmatic LLM prompt structure that utilizes

the available operations in the environment and the specification of similar programs for objects, so that the plan generation can span the environment, robot capabilities, and tasks. In addition, LLM is often combined with the Planning Domain Definition Language (PDDL) to improve the planning performance of the LLM planner. Silver et al. [58] directly provided the planning problem in the PDDL syntax to LLM to generate action sequences. Although the planning performance has been improved, this method requires additional knowledge to build a PDDL file. Liu et al. [71] skillfully used the translation function of LLM to convert natural language into PDDL problem, and then solved it by traditional planner, and finally translated the solution back to natural language. Similarly, Tomoya et al. [72] transform natural language tasks into symbolic sequence representations by using LLM, and then derive the best task program by performing task planning based on Monte Carlo Tree Search (MCTS).

In the realm of complex long-horizon planning, researchers have explored several strategies to enhance the capabilities of LLM-based robotic systems. One approach involves hierarchical and iterative frameworks. For instance, Chen et al. [73] proposed the Hierarchical Multiscale Diffuser (HM-Diffuser), which employs a hierarchical structure to train on trajectories extended at multiple temporal scales. This is complemented by Progressive Trajectory Extension (PTE), an augmentation method that iteratively generates longer trajectories by stitching shorter ones, thereby efficiently managing tasks across different time horizons. Another direction is the integration of explicit planning modules. Erdogan et al. [74] introduced the Plan-and-Act framework, which decouples the planning process into two distinct components: a Planner model that generates structured, high-level plans to achieve user goals, and an Executor model that translates these abstract plans into environment-specific actions. To improve the quality of these plans, their framework also incorporates a scalable synthetic data generation method. Furthermore, a widely adopted strategy to mitigate the complexity of long-horizon tasks is hierarchical task decomposition. This approach breaks down a complex mission into a sequence of manageable sub-goals. For example, Wei et al. [56] leveraged chain-of-thought (CoT) prompting to elicit step-by-step reasoning, enabling the model to tackle complex tasks more effectively. Building on this, Liu et al. [69] proposed DELTA, which decomposes long-horizon objectives into an autoregressive sequence of sub-goals for automated planners. Similarly, Cao et al. [75] developed the LLM-Collab framework, which systematically divides complex planning into four stages-analysis, planning, verification, and improvement-each further decomposed into sub-tasks. Zhen et al. [76] merged human expertise with LLMs using a specialized “Think Net Prompt” and a hierarchical decomposition

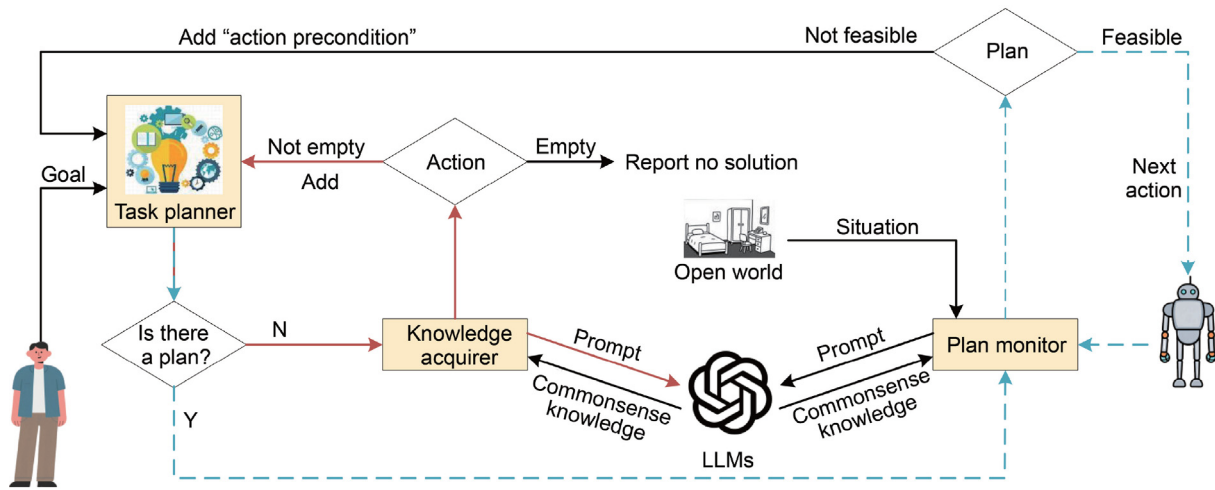


Fig. 5. Schematic diagram of the architecture of COWP. Source: Adapted from [80].

strategy, while Kannan et al. [70] employed LLMs to decompose tasks for multi-robot collaboration, addressing the limitations of single-function robots.

However, the possibility of hallucinations in the output of LLM will accumulate due to the increase of planning period. Therefore, in complex long-horizon task planning, how to suppress or reduce the generation of hallucinations is a major difficulty in using LLM for task planning. Ren et al. [68] proposed KnowNo, a framework for measuring and adjusting the uncertainty of LLM-based planners. KnowNo is based on conformal prediction theory and provides statistical assurance for mission completion while minimizing manual help in complex multi-step planning settings. Park et al. [77] designed a LLM uncertainty estimation method to classify whether the command is certain or uncertain. After determining the uncertainty, it classifies the input as certain or uncertain according to a predefined threshold. Once the command is classified as an uncertain command, the LLM generation problem is used to interact with the user to eliminate the ambiguity of the command. Ong et al. [78] consider the uncertainty in planning by combining a simple method, which emphasizes quantifying uncertainty and exploring alternative paths for task execution. By setting an appropriate probability threshold in skill selection, a method of measuring uncertainty is established to select a better path to perform tasks. Wang et al. [79] proposed a task planning method based on constrained LLM prompt scheme, which can generate executable action sequences from commands. Furthermore, a special processing module is proposed to deal with the LLM illusion problem, so as to ensure that the results generated by LLM are acceptable in the current environment.

4.2. Compensation technology of environmental perception defects

Although the text-based LLM task planning method has made great progress, it still has an inherent limitation, that is, the perception of the physical environment. The input form of pure text makes researchers have to study compensation strategies to deal with this challenge. The current research mainly solves this problem through static knowledge injection [81].

Static knowledge injection is shown in LLM-grop [67]. Before the task planning of desktop object placement, the current environmental information, including known items such as knives, forks, dinner plates, and their existing states, is pre-entered into LLM. Combined with the common sense knowledge of LLM in the form of knowledge base, task planning is performed instead of environmental awareness. Mu et al. [82] introduced KGGPT, an innovative system that integrates prior knowledge.

This system operates by extracting pertinent information from a knowledge graph, transforming it into semantic representations, and then linking these to ChatGPT. KGGPT leverages knowledge graphs to encode robotic skills, task regulations, and environmental limitations. By doing so, it effectively bridges the gap between ChatGPT's existing knowledge base and the practical requirements of real-world service environments. This task planning assumes the invariance and stability of the state of each object in the task space, but in a family environment where dynamic changes may occur, a predefined knowledge base often results in inaccurate or unexecutable task planning results. Dynamic state update better handles the situation where the state of the object may change in an open home environment. Ding et al. [80] developed COWP, an open-world robot planning system that ingeniously integrates a traditional knowledge-based task planning framework with a pre-trained language model, as shown in Fig. 5. The latter serves as a powerful means for acquiring domain-agnostic common-sense knowledge. This synergy endows COWP with the unique ability to apply universal common-sense knowledge to specialized task-planning scenarios. As a result, COWP can effectively handle unanticipated situations that robots may face during the planning process, enhancing their adaptability in open-ended and dynamic environments. However, this ability to encounter and deal with emergencies is precisely what the large language model of plain text input cannot achieve.

5. Vision-based LLM task planning

The inherent limitations of text-based LLM planners in spatial perception and dynamic adaptation have promoted the rise of VLMs and regarded them as a transformative method in the field of robot task planning. By constructing a unified visual-linguistic representation space, VLM effectively bridges the perception-reasoning gap in a pure LLM system [83,84]. In the context of service robot task planning, the VLM-based method presents a diversified development trend. In addition, the introduction of visual perception has also stimulated researchers' exploration in active task cognition. In this section, we will explore the research progress of VLM-based task planning methods and active task cognition.

5.1. Methodological and frameworks innovations

Integrating VLMs into the field of service robot task planning is a major improvement on the traditional text-based method. This part deeply studies the methods and frameworks of using

Table 3
Typical literature on methodological and frameworks innovations.

Paper	Year	Core innovation	Limitations
Li et al. [66]	2024	Aligns visual attributes with LLM via VLM for precise plans, enhanced by ontological knowledge	Requires full attribute labeling and predefined ontologies; struggles with novel objects
Tang et al. [85]	2025	Automates 3D reconstruction from 2D inputs, enhancing 3D scene understanding for dynamic environments	Accuracy drops in sparse or occluded scenes and requires pre-trained VLM with 2D grounding
Ni et al. [86]	2024	Graph networks model object relations for semantic reasoning, while LLM enables natural instruction decomposition	Graph construction becomes costly in cluttered scenes and limited to pre-defined relationship types
Ekpo et al. [87]	2024	Scene graphs enable explicit spatial relationship verification and iterative LLM plan correction ensures physical enforceability	Graph construction latency and fails on ambiguous spatial queries
Mei et al. [88]	2024	Dual-loop correction improves error recovery robustness and maintains plan consistency during dynamic environment changes	Iterative corrections increase task completion time and dependent on VLM's visual grounding accuracy
Wu et al. [89]	2023	Tight LLM-VLM alignment ensures physically feasible plans and context-aware constraint handling	Requires complete 3D object models for constraint verification and struggles with implicit physical rules

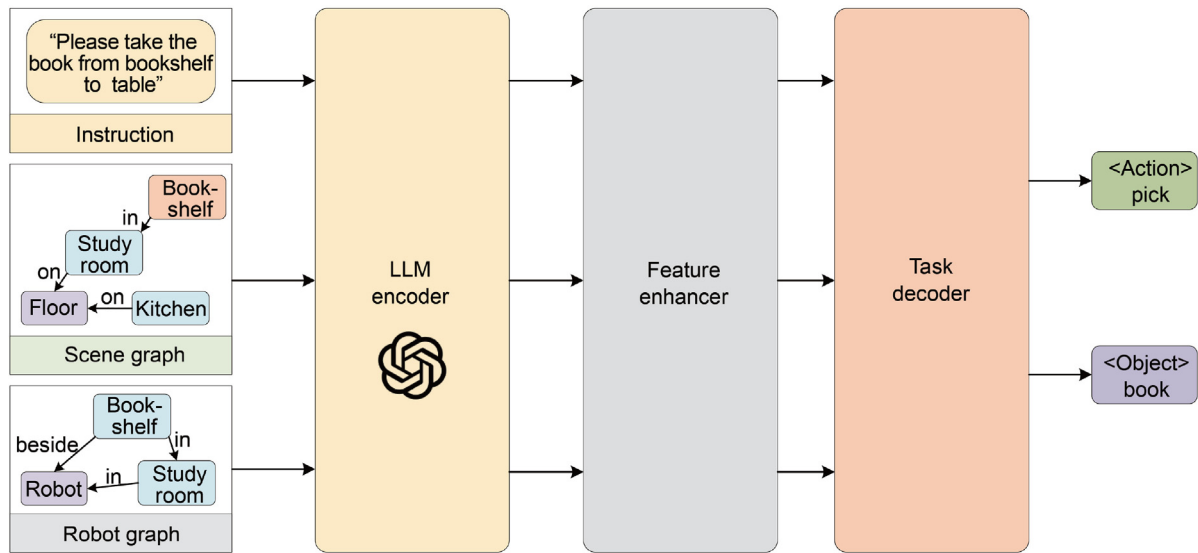


Fig. 6. The architecture diagram of GRID. Source: Adapted from [86].

VLMs to enhance the spatial perception and dynamic adaptability of service robots. By bridging the gap between visual perception and language understanding, VLMs enable robots to interpret complex environmental cues and perform tasks with higher accuracy and flexibility. This section will introduce Text-based LLM Planning in terms of the complexity of instructions, and some of the representative work is shown in Table 3.

In spatial representation and semantic comprehension, Huang et al. [90] proposed VLMs, which is a spatial mapping representation that directly combines pre-trained visual language features with 3D reconstruction of the physical world. In this way, service robots can position space targets outside object-centric space targets during mission planning and execution, such as 'between the TV and the couch'. At the same time, Tang et al. [85] proposed a new framework that uses a 2D prompt synthesis module to enable VLM to train on 2D images and texts, and can independently extract accurate 3D spatial information without manual intervention, thereby significantly enhancing 3D scene understanding, enabling robots to plan and perform actions adaptively in a dynamic environment.

For task decomposition and planning refinement, Ni et al. [86] proposed a new method called graph-based robot instruction decomposer (GRID), as shown in Fig. 6. This method uses the LLM and graph attention networks to encode the object attributes and relationships in the graph, so that the robot can obtain the semantic knowledge widely observed in the environment from

the scene graph. Ekpo et al. [87] proposed VeriGraph, which is a new framework for integrated VLM for robot planning. VeriGraph uses scenario graphs as intermediate representations to capture key objects and spatial relationships, thereby improving planning validation and optimization. The system generates a scene map from the input image and uses it to iteratively check and correct the sequence generated by the LLM-based task planner to ensure compliance with constraints and enforceability.

Regarding dynamic adaptation and self-correction, Mei et al. [88] proposed a ReplanVLM framework for robot task planning, as shown in Fig. 7. An internal error correction mechanism and an external error correction mechanism are proposed for error correction at the corresponding stage. Shirai et al. [97] proposed a visual language interpreter ViLaln, which is a new problem description framework generated by the most advanced LLM and visual language model. It can optimize the generated problem description through the error message feedback of the symbol planner.

In embodied task execution and physical constraint management, Wu et al. [89] developed a TaPA for landing planning with physical scene constraints in embodied tasks. The agent generates an executable plan based on the objects in the scene by aligning the LLM with the visual language perception model. Li et al. [66] proposed the fine-grained task planning FGTP, which aligns the object attributes with their corresponding images through VLM to obtain the actual attributes of the object. This is a method

Table 4
Typical literature on visual reasoning.

Paper	Year	Core innovation	Limitations
Mo et al. [91]	2021	Proposed a graph-based multimodal framework for joint learning and segmentation	Requires paired multimodal data; sensitive to modality misalignment
Liu et al. [92]	2021	Replaces bounding-box features with fine-grained region attention; more precise relationship prediction in scene graphs	Higher computational cost than bounding-box approaches and needs detailed region-level annotations for training
Zhang et al. [93]	2022	Novel 3D scene graph structure combining geometric and semantic information	Real-time graph updates may be challenging in dynamic environments
Xie et al. [94]	2022	Integrates visual concepts with external knowledge bases; generates semantically rich questions about images	Relies on manually curated knowledge bases; limited to domains covered by the knowledge base
Hao et al. [95]	2024	Chain-of-Thought enhanced spatial reasoning; improves SQA accuracy for complex spatial queries	Dependent on LLM reasoning reliability; limited to predefined spatial primitives
Cheng et al. [96]	2024	Specialized in geometric relationship reasoning; bridges visual perception with symbolic reasoning	Prone to LLM hallucination in spatial descriptions; requires accurate object detection as input

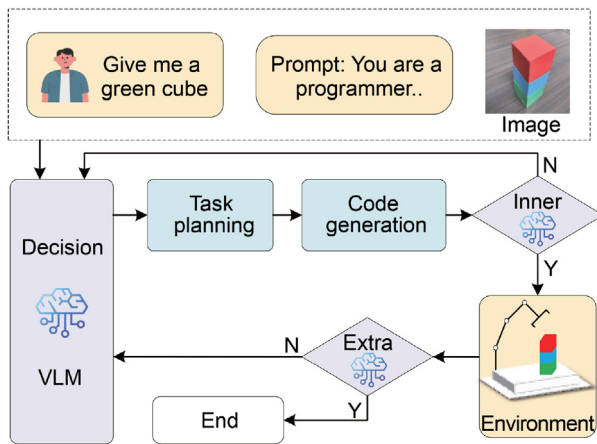


Fig. 7. The framework of ReplanVLM. Source: Adapted from [88].

that combines object ontology knowledge with the LLM to create action sequences. In order to deal with the difficulties encountered in dealing with complex tasks and interacting with the environment effectively, Luan et al. [98] introduced VLM as an environment-aware sensor and assimilated the results into LLM, thus integrating task objectives with environmental information and enhancing the enforceability of task planning output. Huang et al. [99] used the combination of VLM and LLM to perform the task of robot-to-human handover with semantic object knowledge, which enhanced the perception and interaction of robots in a dynamic environment and paved the way for more seamless human-machine collaboration.

Furthermore, Zhang et al. [100] developed GPTArm and proposed a robot task processing framework RTPF, which integrates real-time visual perception, situational reasoning and autonomous strategy planning, so that the manipulator can interpret natural language commands, decompose user-defined tasks into executable sub-tasks, and dynamically recover from errors. Wake et al. [101] proposed a ready-to-use multi-mode task planner using VLM and LLM, and introduced a pipeline to enhance the general visual language model GPT-4V to promote the one-time visual teaching of robots, successfully transforming human actions in videos into robot executable programs. Liu et al. [102] proposed a VLM-driven method to understand scenes in unknown environments. The visual language model is used as the skeleton of the language model for image description and scene understanding, which further proves that the task planning ability of the service robot in the home environment is significantly improved through the visual language model.

5.2. Active task cognition

Active task cognition necessitates that robots possess the capability to comprehend their environment and autonomously deduce task-related information. Fundamentally, active task cognition can be characterized as a visual reasoning task, wherein the reasoning process serves as the cornerstone of the cognitive procedure. This form of visual reasoning extends beyond mere object identification within a scene [103]; it encompasses an understanding of the interactions among these objects, as well as their respective significance and function in the real world. This section elaborates on active task cognition from visual reasoning, which is typical as shown in Table 4.

In the realm of visual cognition research, an emerging body of literature underscores the criticality of representing relational structures and semantic information gleaned from images. Mo et al. [91] introduced a graph learning-based multimodal prior guided segmentation framework, which innovatively addresses the dual challenges of feature extraction and cross-modal correspondence. This framework enables the efficient extraction of modality-specific features while establishing robust regional correspondences across multiple modalities, thereby facilitating more comprehensive visual understanding. Yang et al. [104] proposed an alternative approach centered on emotional reasoning. Their methodology involves constructing an emotional graph that integrates semantic concepts and visual features. Leveraging Graph Neural Network (GCN) techniques, they perform reasoning operations on the emotional graph, ultimately generating emotion-enhanced object features. These advancements collectively highlight the potential of graph-based models in enriching visual cognition by encoding complex semantic and relational information.

In addition, integrating the scene graph into the intelligent system not only enhances the visual cognition, but also emphasizes the relationship and attributes of a single object, which promotes the scene understanding [105–107]. Liu et al. [92] devised a region-aware attention learning approach. This method emphasizes fine-grained visual regions over coarse-grained bounding box features, thus optimizing the image scene graph generation process. Meanwhile, Zhang et al. [93] innovatively incorporated cooking logic into food image component recognition. By extracting semantic information from images, their method directly generates corresponding recipes and preparation techniques, bridging visual understanding and practical application. Jiao et al. [108] planned and designed a 3D scene graph representation for the robot sequence by abstracting the scene layout of robot-scene interaction. These models can effectively capture the spatial and semantic relationships of objects to promote the robot's cognition of tasks.

In addition, the application of Visual Question Answering (VQA) technology also contributes to enhancing the extraction of

semantic information. Frankli et al. [109] proposed RobotVQA, an innovative robot vision architecture designed for VQA tasks. The architecture uses RGB or RGBD images of the scene where the robot is located as the original input, and accurately identifies related objects in the scene through advanced object detection algorithms. On this basis, the semantic map of the scene is systematically constructed by analyzing the qualitative spatial relationship, so as to provide structured environmental information support for subsequent intelligent decision-making and task execution. Xie et al. [94] proposed a knowledge-based VQA model, which innovatively integrates visual concepts and non-visual knowledge to generate deep semantic problems for images, and significantly improves the scene cognition level of robots. Similarly, Hao et al. [95] proposed a novel spatial reasoning paradigm, which significantly improves the performance of Situational Question Answering (SQA) system in spatial relationship reasoning and complex query processing by organically integrating basic model and thinking chain mechanism. This paradigm enables the system to analyze spatial semantic information more accurately through a structured reasoning process. In order to help robots understand scene context information, Wang et al. [110] constructed a chain guided learning model for graphical question answering (DQA). The model relies on the LLM to guide the graph analysis tool, which significantly improves the accuracy of graph analysis and enriches the background knowledge system. Coincidentally, Cheng et al. [96] proposed spatial region GPT (SpatialRGPT), which is dedicated to spatial perceptual VQA. By strengthening spatial perception and reasoning ability, it can effectively promote the cognitive process of robot tasks.

In the above research, active task cognition mainly focuses on three technical directions: Graph-based Reasoning Systems: Several studies [91–93,108] have developed sophisticated graph representations that enable robots to actively reason about object relationships and task contexts. For instance, Jiao et al. [108] proposed a 3D scene graph representation that abstracts scene layouts and robot-scene interactions, allowing robots to dynamically update their understanding of the environment during task execution. VQA Frameworks: Advanced VQA systems [94,95,109,110] empower robots to actively query their visual understanding. The RobotVQA system [109] demonstrates how robots can construct semantic maps by analyzing qualitative spatial relationships through a question-answering paradigm. Multimodal Chain-of-Thought Reasoning: New frameworks like SpatialRGPT [96] combine visual perception with step-by-step reasoning, enabling robots to actively verify their understanding through multimodal evidence. Despite these advances, current systems still face limitations in genuine active cognition. Most approaches remain reactive rather than proactive, requiring explicit human queries or predefined triggers rather than autonomously identifying task-relevant information. At present, how to construct a reasoning model with active task cognitive ability based on VLM and visual perception information has become a key challenge to be overcome.

6. Audio-based LLM task planning

Although visual and textual modalities dominate in robot task planning, audio signals provide a unique channel for the perception of service robots, especially in visual deprivation scenarios (e.g., dark environments or assisted care). Audio-based LLM mission planning leverages voice commands, ambient sound, and acoustic cues to enable robots to interpret implicit user intents and dynamic contexts. However, compared with the vigorous development of LLM task planning based on text and visual input, audio-driven task planning still faces many challenges, and related research is in a stage of rapid development. This part studies

how LLMs handle audio input for task planning, and elaborates on the work of Audio Language Model (ALM), acoustic scene perception and speech interaction. Typical examples are shown in Table 5.

In terms of the basic model architecture, several breakthrough studies have laid a solid foundation for the joint representation of audio and language. As shown in Fig. 8, the CLAP (Contrastive Language-Audio Pretraining) model proposed by Elizalde et al. [111] effectively connects the language and audio modalities through a dual-encoder structure and a contrastive learning objective, mapping audio and text descriptions into a joint multimodal space to enable flexible class prediction during inference and significantly enhance cross-modal retrieval performance. Building on this work, the team led by Guzhov [112] integrated the high-performance audio model ESResNeXt with the text-image contrastive model CLIP, as shown in Fig. 9, constructing a three-modal hybrid architecture that excels in environmental sound classification tasks and extends the zero-shot learning capability of the basic model to the audio domain. Furthermore, Wu et al. [119] proposed Wav2CLIP, which innovatively projects audio, images, and text into the shared embedding space of CLIP, offering an efficient solution for tasks like zero-shot sound classification and demonstrating strong adaptability and generalization in multimodal scenarios, thus advancing the field of audio-language joint representation.

To bolster models' comprehension of intricate audio scenes, researchers have engineered diverse specialized architectures. The Audio Flamingo model, devised by Kong et al. [114], exhibits remarkable few-shot learning and conversational prowess, enabling it to rapidly adapt to unseen task scenarios. Gong et al. [120] introduced LTU (Listen, Think, and Understand), a novel audio foundation model that demonstrates robust performance and generalization on traditional audio tasks, including classification and captioning. Notably, LTU showcases emergent audio reasoning and understanding capabilities hitherto absent in existing models. As shown in Fig. 10, Ghosh et al. [113] proposed GAMA, a general-purpose large audio language model that integrates advanced audio comprehension and sophisticated reasoning abilities, thereby offering a novel paradigm for the semantic analysis of ambient sounds.

In the realm of voice interaction, numerous studies have focused on enhancing robots' ability to understand and execute spoken instructions. To handle the subtleties of oral commands in scenarios like social navigation, Sun et al. [121] proposed "Beyond Text", a method that improves LLM decision-making by integrating audio transcription with functional sub-parts, seamlessly combining text-based guidance with a human-auditory information language model and marking a significant advancement in social robot navigation and broader human-robot interaction [122]. Meanwhile, Chen et al. [115] introduced a novel Speech Augmented Language Model (SALM), featuring a frozen text LLM, an audio encoder, a modality adapter module, and LoRA layers to adapt to voice input and task instructions. The unified SALM not only achieves performance comparable to the Conformer baseline for automatic speech recognition (ASR) and speech translation (AST) tasks but also demonstrates zero-shot context learning capabilities, highlighting its adaptability in voice interaction scenarios. Moreover, Rubenstein et al. [116] introduced AudioPaLM, a sophisticated large language model designed for speech understanding and generation. AudioPaLM integrates the text-based language model PaLM-2 [123] and the speech-based language model AudioLM [124] into a unified multimodal architecture. This innovative integration endows AudioPaLM with the ability to process and generate both text and speech, enabling it to perform tasks such as speech recognition and speech-to-speech translation with remarkable efficiency and accuracy.

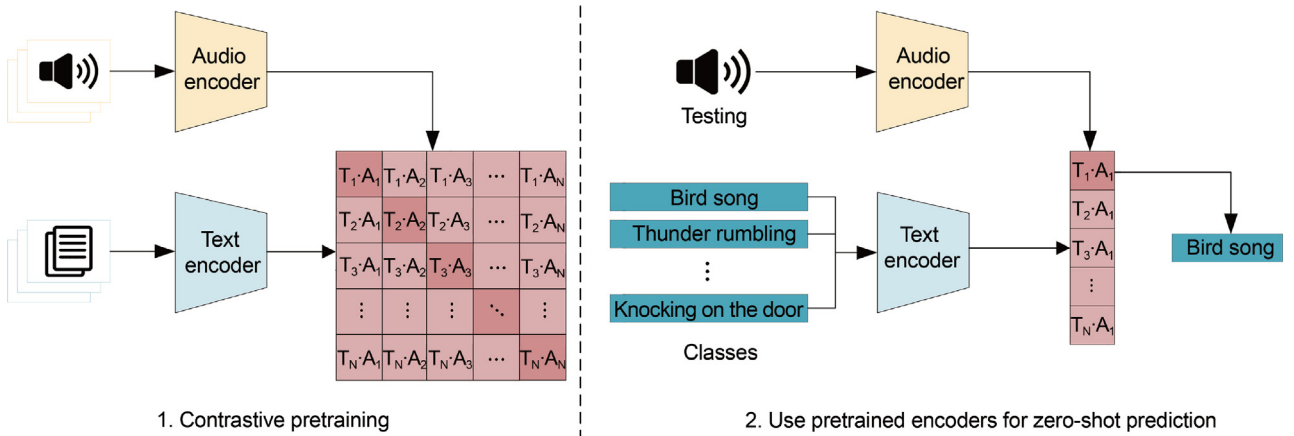


Fig. 8. The architecture diagram of CLAP. Source: Adapted from [111].

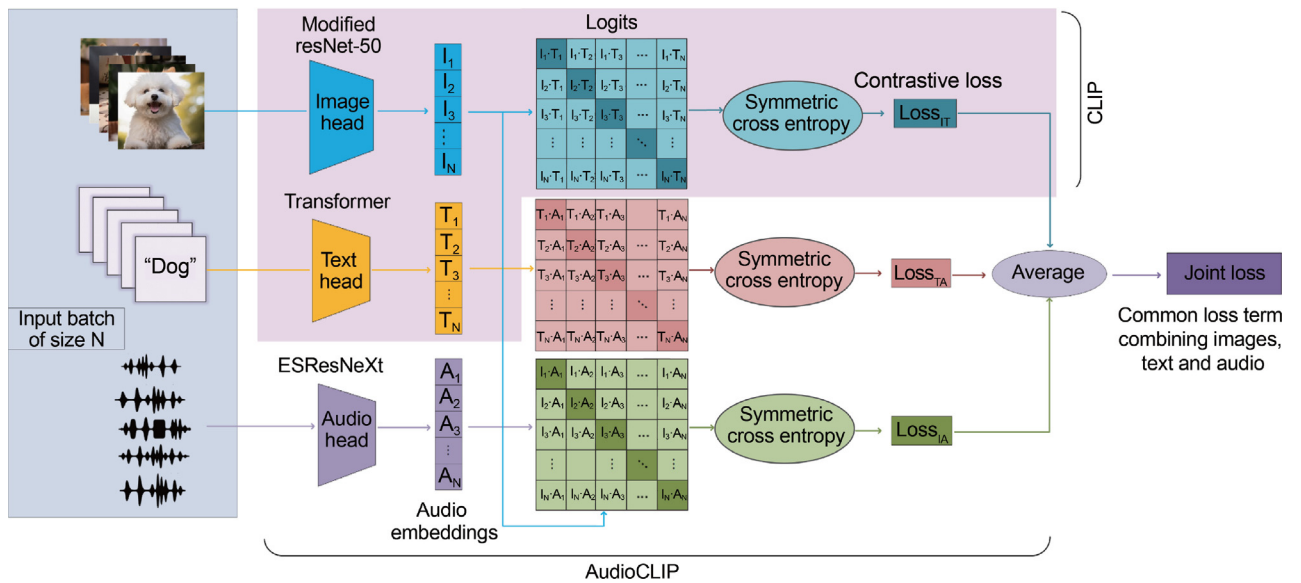


Fig. 9. Workflow of the proposed AudioCLIP model. Source: Adapted from [112].

Table 5

Typical literature on audio-based LLM task planning.

Paper	Year	Core innovation	Limitations
Elizalde et al. [111]	2023	Dual-encoder contrastive learning for audio-text alignment; enables zero-shot audio classification and retrieval	Requires large-scale aligned audio-text pairs; struggles with fine-grained audio semantics
Guzhov et al. [112]	2022	Tri-modal fusion via CLIP extension; ESResNeXt audio encoder for environmental sound classification	Limited to coarse sound categories; no joint training of all modalities
Kong et al. [114]	2024	Few-shot learning for unseen audio tasks; conversational audio reasoning via LLM integration	Context window limits long audio processing; overfits to frequent sound patterns
Chen et al. [115]	2024	Maintains LLM knowledge while adding audio via adapters; Zero-shot adaptation to new voice instructions	Audio encoder limits real-time processing; degrades with noisy speech
Rubenstein et al. [116]	2023	Unified PaLM-2 + AudioLM architecture; bidirectional speech-text generation	Relies on high-quality parallel speech-text corpora; generated speech lacks emotional nuance
Wang et al. [117]	2024	Gradient-free adaptation to speakers; in-context learning with few audio-text examples	Requires representative few-shot examples; limited to vocabulary seen in context
Deshmukh et al. [118]	2023	Unified text-generation framework for all audio tasks; eliminates task-specific fine-tuning	No explicit audio-text alignment; struggles with novel sound events

To assess the performance of audio comprehension models in tasks demanding expert-level knowledge and intricate reasoning, Sakshi et al. [125] introduced MMAU (Multimodal Multilingual Audio Understanding), a benchmark featuring 10,000 meticulously curated samples along with manually annotated natural

language questions and answers, covering speech, environmental sounds, and music. In a different yet equally impactful vein, Wang et al. [117] proposed a novel Speech-based In-context Learning (SICL) approach that harnesses a few context examples—paired spoken words and labels—from specific dialects or speakers to

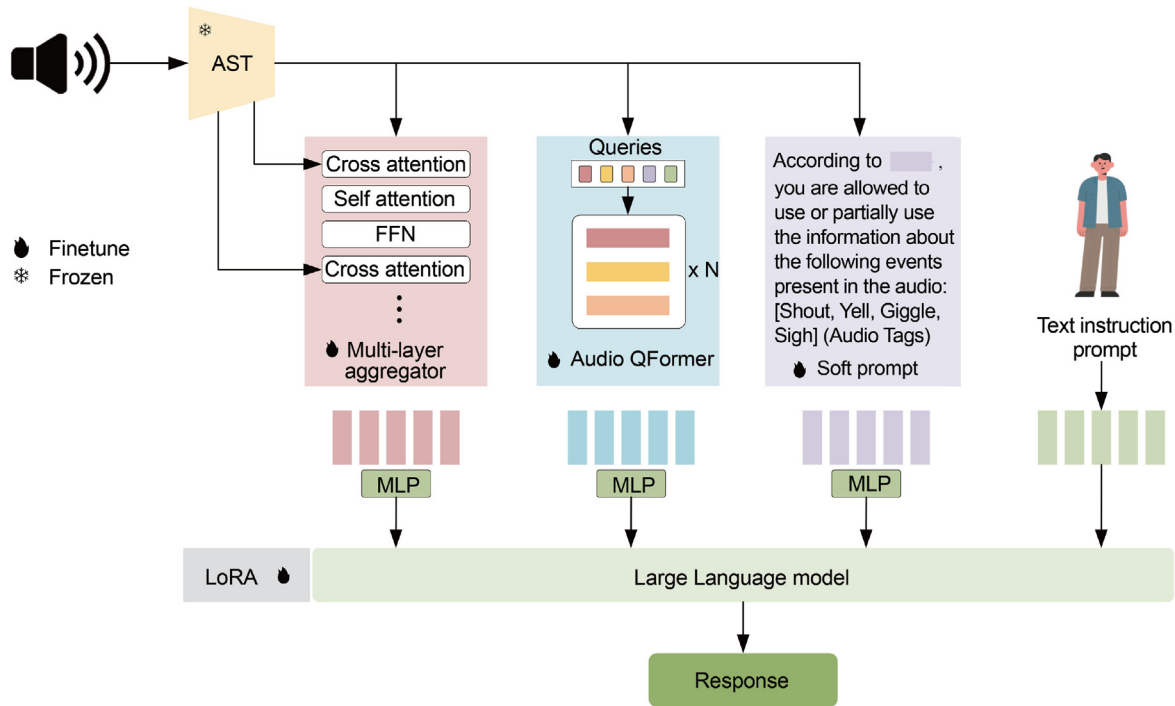


Fig. 10. The schematic diagram of GAMA. Source: Adapted from [113].

achieve language- or speaker-level model adaptation during testing without gradient descent, effectively resolving the adaptability issues stemming from dialect and speaker disparities. At the application level, Salewski et al. [126] developed ZerAuCap, a model guided by a pre-trained audio language model to generate captions that accurately describe audio content. It utilizes audio context keywords to prompt the language model, enabling the generation of text closely associated with the sounds. Meanwhile, addressing the limitation in generating appropriate language for open-ended tasks like audio captioning and audio Q&A, Deshmukh et al. [118] introduced Pengi. This innovative model capitalizes on transfer learning by formulating all audio tasks as text generation tasks. Pengi's unified architecture can seamlessly handle both open and closed tasks, eliminating the need for additional fine-tuning or task-specific modifications, thus offering a versatile and efficient solution for various audio-language processing needs.

Despite these advances, audio LLM planning still faces many challenges. The ClozeGER error correction framework proposed by Hu et al. [127] for speech recognition errors improves the fidelity of the corrected output by introducing a multimodal LLM (SpeechGPT) to receive the source speech as additional input, but its robustness in complex noise environments still needs to be improved.

7. Multimodal-based LLM task planning

With the rapid development of LLM, limited by its narrow sensory input and fragile environmental understanding ability, people are no longer satisfied with a single-modal LLM, but try to integrate the input of multiple modalities [128] and integrate visual, language, audio and other signals into a MLLM. As a transformative paradigm of robot planning, MLLM enables robots to understand fuzzy instructions, adapt to dynamic environments, and learn from cross-modal correlations. In this section, we will introduce the current research progress and application of multimodal large language models.

7.1. Multimodal large language models

In this section, we will explore a series of typical MLLMs and analyze the differences between them. The Flamingo architecture proposed by Alayrac et al. [129] firstly used the query-based cross-attention mechanism, that is, the 'sensor resampler'. As shown in Fig. 11, the architecture connects the powerful pre-trained pure visual model with the pure language model, and can handle any alternating visual and text data sequences, as shown in Fig. 12. This innovative way to build a strong visual language interaction module, showing the excellent ability to learn in the context of the environment. Similarly, BLIP-2 [130] uses a similar method in the image coding process, using the Qformer model to extract image features. Subsequently, the model promotes the interaction between image and text through the cross-attention mechanism. When fine-tuning for a specific downstream task data set, BLIP-2 will unlock the visual encoder and fine-tune it together with Qformer.

In addition, Yang et al. [131] proposed MM-REACT, which combines ChatGPT with various visual models to complete multimodal tasks. The system is similar to the previous research results using the visual question answering caption model and the language image model. However, the special feature of MM-REACT is that it has the ability to independently determine whether to call the visual model. The LLaMA-Adapter proposed by Zhang et al. [132] realizes an efficient fine-tuning process through an adapter, thus expanding to multimodal application scenarios. MiniGPT-4 proposed by Zhu et al. [133], combines BLIP-2 and Vicuna, and uses a projection layer to align the frozen visual encoder with the frozen high-level large language model Vicuna, thereby achieving advanced multimodal functions. LLaVA [134] uses GPT-4 to generate command fine-tuning data, and further strengthens the command tracking ability of the multimodal large language model by training the visual command adjustment data. LLaVA-1.5 [135] is extended on this basis, and the VQA data set is integrated into the instruction adjustment data, which greatly improves the performance in a series of benchmark tests.

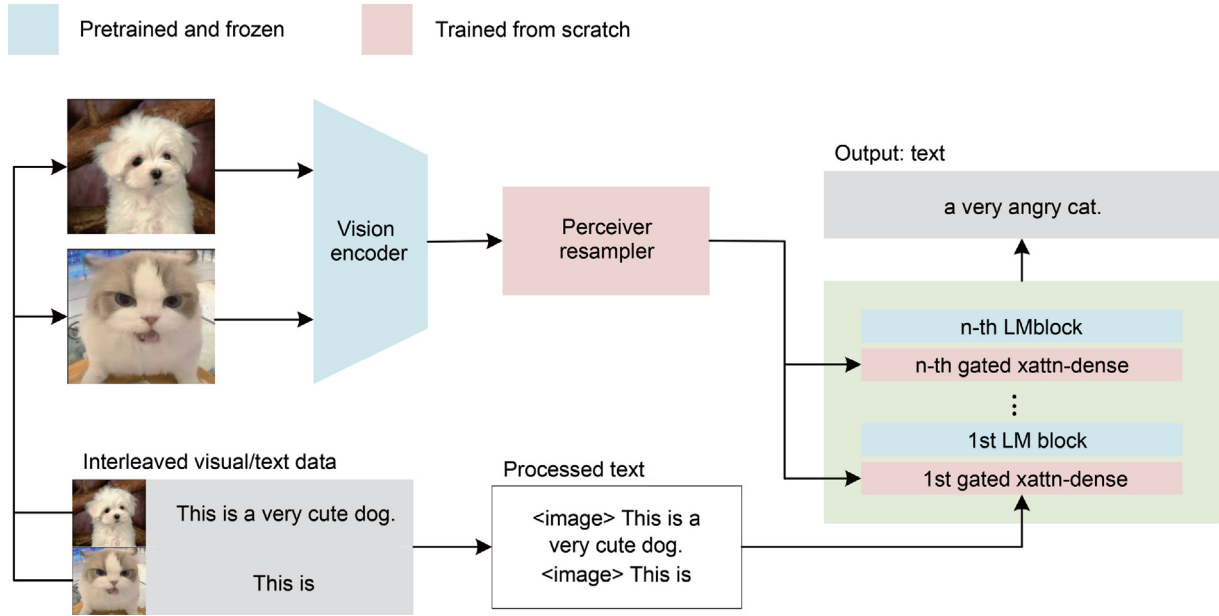


Fig. 11. The architecture diagram of Flamingo. Source: Adapted from [129].

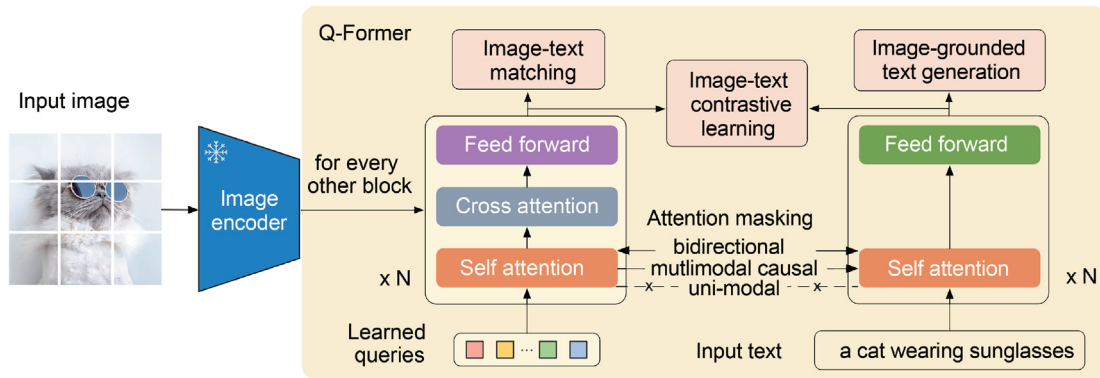


Fig. 12. The schematic diagram of BLIP-2. Source: Reproduced with permission from [130]. © 2023 JMLR.org.

Kosmos-2, introduced by Peng et al. [136], implements new capabilities for perceiving object descriptions (i.e., bounding boxes) and anchoring text to visual scenes by representing reference expressions as links in the Markdown format, where objects are described as sequence of position markers.

In contrast, QWen2-VL [137] adopts a more complex three-stage training process and introduces a Naive Dynamic Resolution mechanism, so that the model can dynamically process images with different resolutions into different numbers of visual markers. This method enables the model to generate more efficient and accurate visual representation, which is closely related to the human perception process. The mPLUG-Owl introduced by Ye et al. [138], through the modular learning of the basic large language model, combined with the visual knowledge module and the visual abstractor module, gives the large language model multimodal capabilities. This method makes full use of the synergy between modalities and improves the performance of plain text tasks and multimodal tasks. At the same time, Otter proposed by Li et al. [139] improved the OpenFlamingo model, which focuses on improving the execution of instructions and effectively uses context samples, showing extraordinary proficiency in multimodal perception, reasoning and situational learning. CogCoM, introduced by Qi et al. [140], introduces ‘Chain of Manipulations’,

which is a mechanism that allows visual language models to solve problems step by step with evidence. After training, the model can solve various visual problems by actively triggering internal operations (such as positioning, magnification) and generating results (such as bounding boxes, images), without the need for external tools, while allowing users to trace the causes of errors. Yan et al. [141] introduced a new framework, Visual Grounding Through Fine-Grained Reward Modeling (ViGoR), which uses fine-grained reward modeling to significantly enhance the visual positioning ability of large-language visual models based on pre-training baselines.

Different modal data have significant differences in terms of presentation, resolution and semantic granularity, and how to unify them to extract common information while retaining their unique details is the core challenge in the fusion process. The above studies have developed a variety of fusion techniques to try to solve these problems, mainly including attention mechanisms, feature alignment, gating mechanisms and graph neural networks. Attention mechanisms have emerged as the predominant fusion strategy in Transformer-based architectures. Cross-attention enables dynamic association between modalities [129], while self-attention captures intra-modal relationships. This dual mechanism allows flexible information interaction tailored to

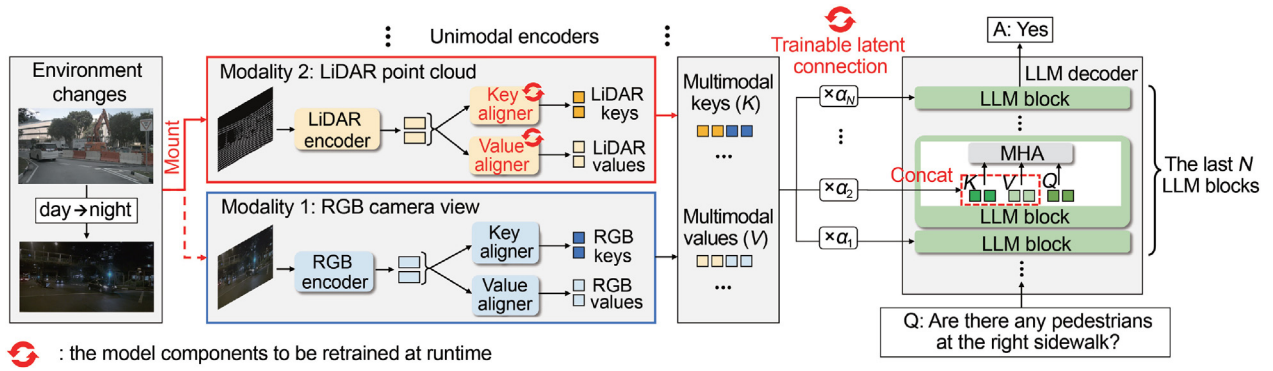


Fig. 13. Example of modal adaptation of mPnP-LLM to two input modes. Source: Reproduced with permission from [143]. © 2023 licensed under CC BY 4.0, ACM.

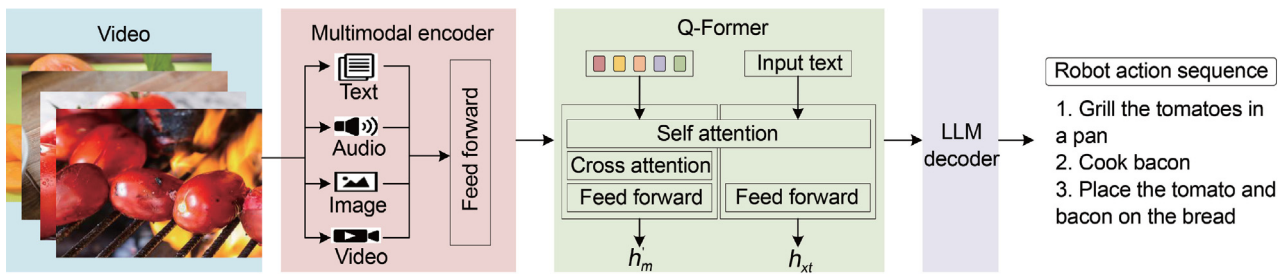


Fig. 14. The schematic diagram of AVBLIP. Source: Adapted from [144].

specific task requirements. Feature alignment techniques create shared latent spaces for heterogeneous data. Contrastive learning methods like CLIP [142] align modalities by maximizing similarity of matched pairs while minimizing unrelated ones. The Q-Former module in BLIP-2 [130] extends this through shared query vectors that learn unified cross-modal representations. Gating mechanisms provide dynamic information flow control. Lightweight adapters, as seen in LLaMA-Adapter [132], enable context-aware feature injection without full model fine-tuning. These systems automatically adjust modality weights based on environmental conditions. Graph Neural Networks offer structural fusion capabilities by representing multimodal elements as graph nodes with spatial/semantic edges. Systems like GRID [86] and Veri-Graph [87] employ Graph Attention Networks to combine visual scene graphs with verbal commands, significantly enhancing spatial reasoning for robotic tasks.

7.2. Technological evolution and practical applications

With the continuous evolution of artificial intelligence models, service robots have become a key area for the application of artificial intelligence technology. With its excellent perception and reasoning efficiency, MLLMs, as the core intelligent center of service robots, play an indispensable role in action command generation and task planning. This section will discuss in depth from multiple dimensions such as model evaluation optimization, robot system integration, and task planning interaction, and systematically analyze the innovative practices and technological breakthroughs of MLLMs in service robot task planning scenarios.

In the field of model evaluation and optimization, researchers are committed to improving the performance and efficiency of MLLMs. The OmniBench benchmark proposed by Li et al. [145] aims to comprehensively evaluate the parallel processing and

reasoning ability of the model for multimodal inputs such as visual, auditory, and text. By constructing a standardized evaluation system, it provides an important basis for quantifying the performance of the model. Aiming at the problem of insufficient multimodal processing efficiency in resource-constrained scenarios, the mPnP-LLM model developed by Huang et al. [143] creatively combines single-modal encoders with dynamically configurable LLM modules and supports dynamic training at runtime, as shown in Fig. 13. This design enables the model to independently select effective modes according to actual needs, which greatly improves resource utilization efficiency. In the face of the application bottleneck caused by the model's dependence on specific input formats, the LLMBind framework proposed by Zhu et al. [146] transforms multimodal input into task-specific token sequences with the help of expert mixing (MoE) mechanism, and realizes efficient processing and output conversion of multimodal data such as image, text, video and audio. In addition, the RespLLM framework proposed by Zhang et al. [147], another way to unify text and audio representations, is applied to respiratory health prediction tasks. The framework makes full use of the rich prior knowledge of pre-trained LLM, and realizes the deep fusion of audio and text information through cross-modal attention mechanism, which provides a new technical paradigm for multimodal model to process heterogeneous data.

At the application level of robot systems, many studies have focused on the deep integration of MLLMs into service robot systems. The service robot system constructed by Ni et al. [148] integrates hardware design, scene modeling algorithm, mobile navigation strategy based on TEB path planning, grasping operation based on pre-training model, and task planning module based on visual language model, forming a complete technical chain and realizing the whole process intelligence from perception to decision-making. The multimodal language model agent

system proposed by Chung et al. [149] combines the task planning function with the interaction ability of the physical robot by integrating the router, the chat robot and the task planner, which significantly improves the decision-making efficiency and interaction flexibility of the robot in practical applications. The research work of Jiang et al. [150] proposes an innovative multimodal prompt formula, which transforms diversified robot operation tasks into a unified sequence modeling problem, and establishes an evaluation protocol including multimodal tasks and system generalization ability, which provides a new direction for the standardization research of robot task planning.

In terms of task planning and environmental interaction, researchers continue to explore how to make robots better adapt to complex environments. The Inner Monolog mechanism proposed by Huang et al. [151], by integrating multimodal environmental feedback information, can generate a task planning scheme that is more suitable for the actual scene, effectively coordinate the input information of different sensors, and enhance the AI's ability to understand and respond to the environment. The system developed by Liu et al. [152] gives full play to the advantages of text understanding and visual processing of MLLMs, and can accurately transform user dialog and environmental visual information into a robot executable operation plan. The framework proposed by Wang et al. [153], with the help of multimodal GPT-4V, deeply integrates natural language instructions with robot visual perception, which significantly improves the accuracy and flexibility of embodied mission planning. The PaLM-E embodied language model proposed by Driess et al. [154] realizes the direct connection between sensor data and language model in the real world through staggered vision, continuous state estimation and text input coding, and builds a bridge between perceptual information and semantic understanding. As shown in Fig. 14, the AVBLIP model proposed by Kambara et al. [144], by introducing a multimodal encoder, supports joint input of video, audio, voice and text, and can efficiently generate robot action sequences. The multimodal interaction framework proposed by Lai et al. [155], integrates voice commands and posture information, combines the global information of the environment obtained by the visual system, uses a large-scale language model to analyze the speech text and bounding box data, and avoids the model to generate unreasonable output through key control syntax constraints, thereby generating reliable robot action sequences. In addition, the service robot task reasoning mechanism proposed by Tian et al. [156], innovatively integrates multimodal information and ontology knowledge, systematically manages user and environment information through ontology knowledge base, collects and integrates multi-modal data such as vision, speech and scene knowledge in real time, and conducts deep reasoning based on fine-tuning LLM, which provides richer and more reliable knowledge support for robot decision-making.

In terms of closed-loop integration, systems like Inner Monolog [149] demonstrate how continuous visual feedback from RGB-D sensors can be integrated with visual-language models to dynamically update task plans when environmental changes are detected. For human-robot collaboration scenarios, approaches such as AudioPaLM [116] enable natural language corrections during task execution, allowing users to verbally adjust the robot's actions when deviations occur. The integration of reinforcement learning from human feedback (RLHF [53]) provides another important mechanism for adapting LLM outputs to real-world uncertainties, where human demonstrations and corrections help align the model's planning with physical constraints and task requirements. These developments highlight the critical need to move beyond static, open-loop task specifications and instead develop systems that can maintain plan feasibility through continuous perception-action cycles and human interaction.

Furthermore, the recent breakthroughs in embodied intelligence, exemplified by Google's Gemini Robotics [157] and NVIDIA's GROOT [158], represent a paradigm shift from traditional LLMs by fundamentally bridging the gap between cognitive AI and physical robotic execution. Unlike conventional LLMs, which are confined to textual data and generate abstract outputs like language or code without direct interaction with the physical world, these novel models are designed as Vision-Language-Action (VLA) systems that integrate perception, understanding, and action into a cohesive whole. Google's Gemini Robotics [157], built upon the Gemini 2.0 model, operates as an on-device VLA architecture, processing real-time sensor inputs such as RGB-D and LiDAR to directly output executable joint control commands, thereby bypassing intermediate symbolic planning and significantly reducing latency for enhanced real-time performance and privacy. Concurrently, NVIDIA's GROOT [158] employs a dual-system architecture where a vision-language module interprets the environment before a diffusion transformer generates fluid motor actions, further distinguished by its use of synthetic data generation through GROOT-Dreams, which compresses skill acquisition from months to 36 h, and its seamless Sim2Real transfer via Isaac Sim 5.0. This architectural and methodological innovation allows both systems to transcend the limitations of traditional cloud-dependent LLMs by embedding physical feasibility checks and uncertainty-aware planning directly into their decision-making processes. The result is an embodied intelligence capable of sensing the physical environment, interpreting high-level task intent from natural language, and translating it into safe, executable physical actions—a critical advancement for applications ranging from industrial automation to service robotics in unstructured environments, thereby tightly coupling the cognitive prowess of AI with the physical dexterity of robotic systems.

8. Current challenges and future prospects

From the review and analysis, it is obvious that significant efforts have been made to improve the planning ability of service robots, and some notable progress has been achieved. However, due to the complexity and diversity of the domestic environment, task planning for service robots still requires substantial exploration. As shown in Fig. 15, the main problems in the field of task planning of service robots based on LLM can be summarized as follows.

- **Insufficiency in real-time performance and safety:** The inference processes of current LLMs often demand considerable computational power and time, thereby limiting the ability to guarantee real-time decision-making. How to realize efficient computation of LLM is a general step towards real-time planning. Compounding this, the inherent latency, coupled with the potential for hallucinations and uncertainties during the planning process, poses increased safety risks for robots during task execution.
- **Challenges of multimodal fusion:** It is still a technical problem to effectively fuse text, vision, speech and other modal information and establish a reasonable fusion strategy. How to design an efficient fusion architecture and balance the weights between different modalities needs further study.
- **Model evaluation and optimization:** The lack of evaluation criteria and tools for multimodal LLM makes it difficult to comprehensively evaluate the performance and robustness of the model. At the same time, how to optimize the model structure and improve the computational efficiency and generalization ability is also a problem to be solved.

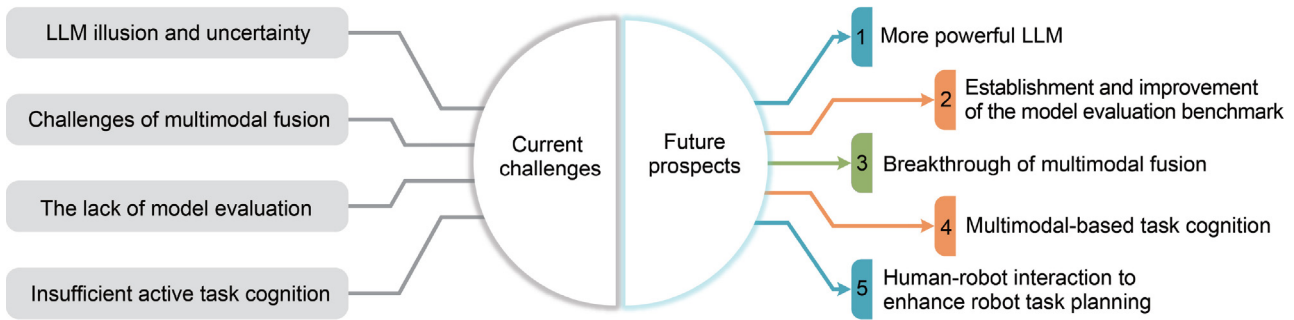


Fig. 15. Some current challenges and future perspectives in task planning for service robots.

- **Insufficient active task cognition:** At present, LLM still has shortcomings in active task cognition. The service robot can only perform task planning and execution according to the user's instructions, and it is difficult to perform autonomous reasoning and planning according to environmental information and task objectives. How to construct a reasoning model with active task cognitive ability needs further exploration.
- **Tactile language models are underdeveloped:** Compared to LLMs, VLMs, and MLLMs, tactile language models remain in their early stages of development. Challenges include scarce tactile datasets, underdeveloped multimodal fusion architectures, and the need for breakthroughs in aligning dynamic contact state representations with language understanding.

In view of the existing problems, the proposed research directions of LLM-based task planning for domestic service robots include:

- **More powerful LLM for robot task planning:** With the continuous development of technology, the performance of LLM has significantly improved [159,160]. In order to enable LLM to respond in real-time [161,162], it is necessary to explore various strategies such as model compression and acceleration techniques [163,164], edge computing deployment [165], and more efficient prompt engineering [35]. Considering the security of task planning, recent studies have also made substantial progress in mitigating hallucinations in LLMs [166]. It is also extremely essential to ensure that the generated plans comply with physical and safety constraints [56,58,66]. To further enhance their generalization ability and robustness in robot task planning and executing service-oriented tasks, it remains essential to incorporate external knowledge bases and complementary techniques [80,167,168].
- **Establishment and improvement of the model evaluation benchmark:** LLMs have been widely used in the field of robotics. Although numerous evaluation benchmarks exist to assess their perception and planning capabilities [169,170], there is a lack of a unified and recognized benchmark for single-modal and multimodal, so that each model data can be quantified. Therefore, it is necessary to establish specific evaluation benchmarks tailored to the task planning field of service robots, such as task success rate, planning efficiency, robustness, etc. This would facilitate a more accurate and comparative assessment of model performance.
- **Breakthrough of multimodal fusion:** The text-based LLM has limitations in spatial perception and dynamic adaptation [71], while visual-based and auditory-based models have difficulties in understanding abstract concepts and extracting effective information in noisy environments, respectively [97,121]. Therefore, the effective fusion of text, visual, auditory and other modal information and the establishment of a reasonable fusion strategy are the key to improving the task planning ability of service robots [129,130,147]. For instance, exploring more efficient modal alignment methods and cross-modal attention mechanisms could enable effective interaction and fusion across different types of modal information [171,172].
- **Multimodal-based active task cognition:** Numerous studies have demonstrated the feasibility of integrating LLM with task cognition [130,173,174]. Through the combination of vision, auditory and LLM, service robots can gain stronger active task cognitive ability, enabling them to make autonomous reasoning and planning according to environmental information, sound signals and task objectives, while adapting to the dynamic changing environments. To this end, further research can be conducted on how to deeply integrate multimodal information with task cognition, such as by developing a unified multimodal task cognition model or incorporating task cognition into the pre-training process of multimodal LLMs.
- **Human-robot interaction to enhance robot task planning:** Human-robot interaction is another critical avenue for enhancing the task planning capabilities of service robots, alongside multimodal information. Beyond perceiving the environment through vision and audio, service robots can infer user intentions through direct dialog and interaction, enabling task planning that better aligns with human preferences [175–177]. By integrating technologies such as speech recognition, natural language processing, and affective computing, more natural and efficient human-robot interactions can be achieved, thereby significantly improving the effectiveness of task planning.
- **Advancing tactile intelligence in robotic systems:** Future research should focus on three key areas: developing comprehensive tactile-linguistic datasets [178], creating efficient architectures for real-time multimodal fusion [179], and advancing closed-loop tactile control in embodied LLMs [180]. These advancements will enable more adaptive robotic interaction through enhanced multimodal understanding and responsive tactile integration, particularly in precision tasks requiring delicate manipulation.

9. Conclusion

This paper systematically reviews the current task planning methods for service robots based on LLMs, with a particular

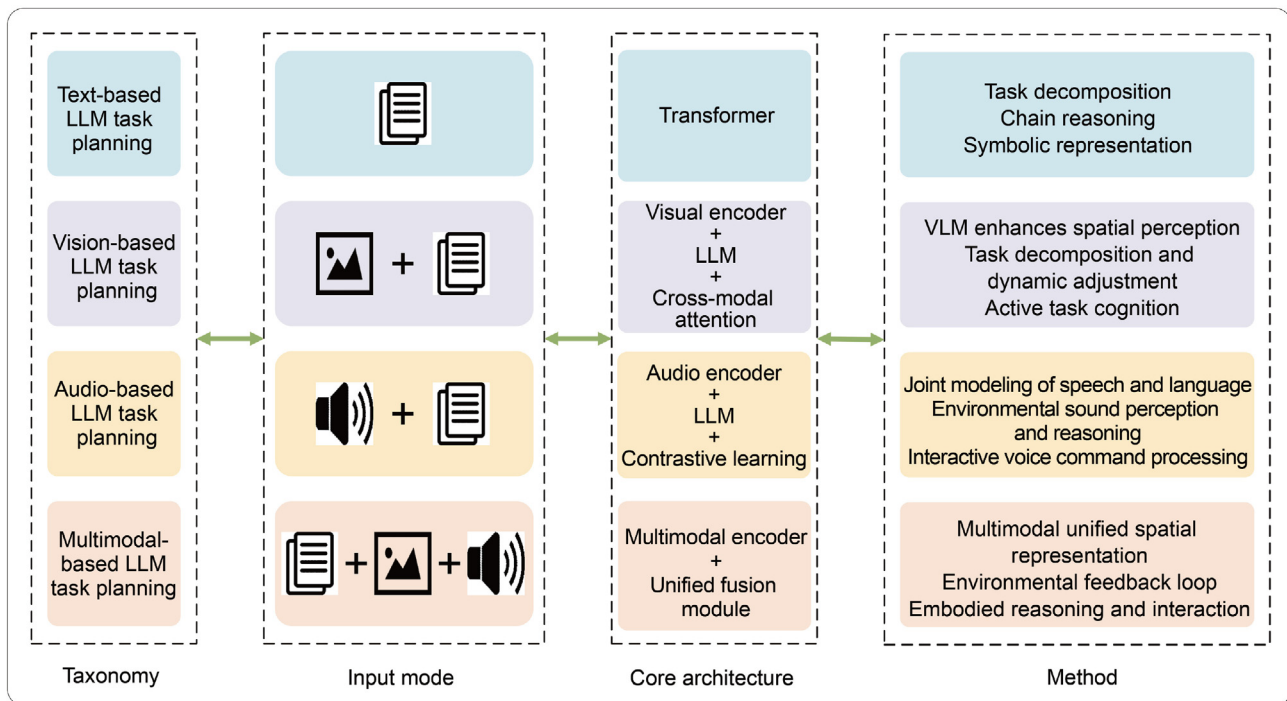


Fig. 16. Task planning for different input modalities.

focus on key challenges and recent advancements. Specifically, the basic knowledge of LLM is introduced. Then, from the unique perspective of input modality difference, the research trends of LLMs based on text, vision, audio and multimodal fusion in task planning are discussed in depth, as shown in Fig. 16. Also, a comprehensive and detailed analysis of the existing systems, frameworks and models is carried out, as well as the bottlenecks and problems faced by current research. Finally, constructive and forward-looking insights are put forward on the future research direction of this field. It is anticipated that the findings of this paper can support the development of more robust, reliable, adaptable, and efficient LLM-based task planning methods for service robots, thereby promoting the field to a new level.

CRedit authorship contribution statement

Shaohan Bian: Writing – review & editing, Writing – original draft, Investigation, Formal analysis. **Ying Zhang:** Writing – review & editing, Writing – original draft, Supervision, Investigation. **Guohui Tian:** Writing – review & editing, Investigation. **Zhiqiang Miao:** Writing – review & editing, Writing – original draft, Investigation. **Edmond Q. Wu:** Writing – review & editing, Investigation. **Simon X. Yang:** Writing – review & editing, Investigation. **Changchun Hua:** Writing – review & editing, Supervision, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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